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Operations Strategy

Lecture 2

Dr Roberto Chavez

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Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.

OUTLINE

- What is operations strategy?
- The strategic role of OM
- Competitive priorities
- The perspectives of OM strategy
- Bridging OM and operational objectives

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WHAT IS STRATEGY?

1. Strategy includes:
   - The position of the organization relative to its environment
     - Where am I now?
   - Long-term objectives
     - Where I want to get?

2. Strategy supports the organization to reach its objectives
   - How to get there?
     - Provide a detailed plan
     - Use appropriate resources and build capabilities

KOMATSU

CATERPILLAR®

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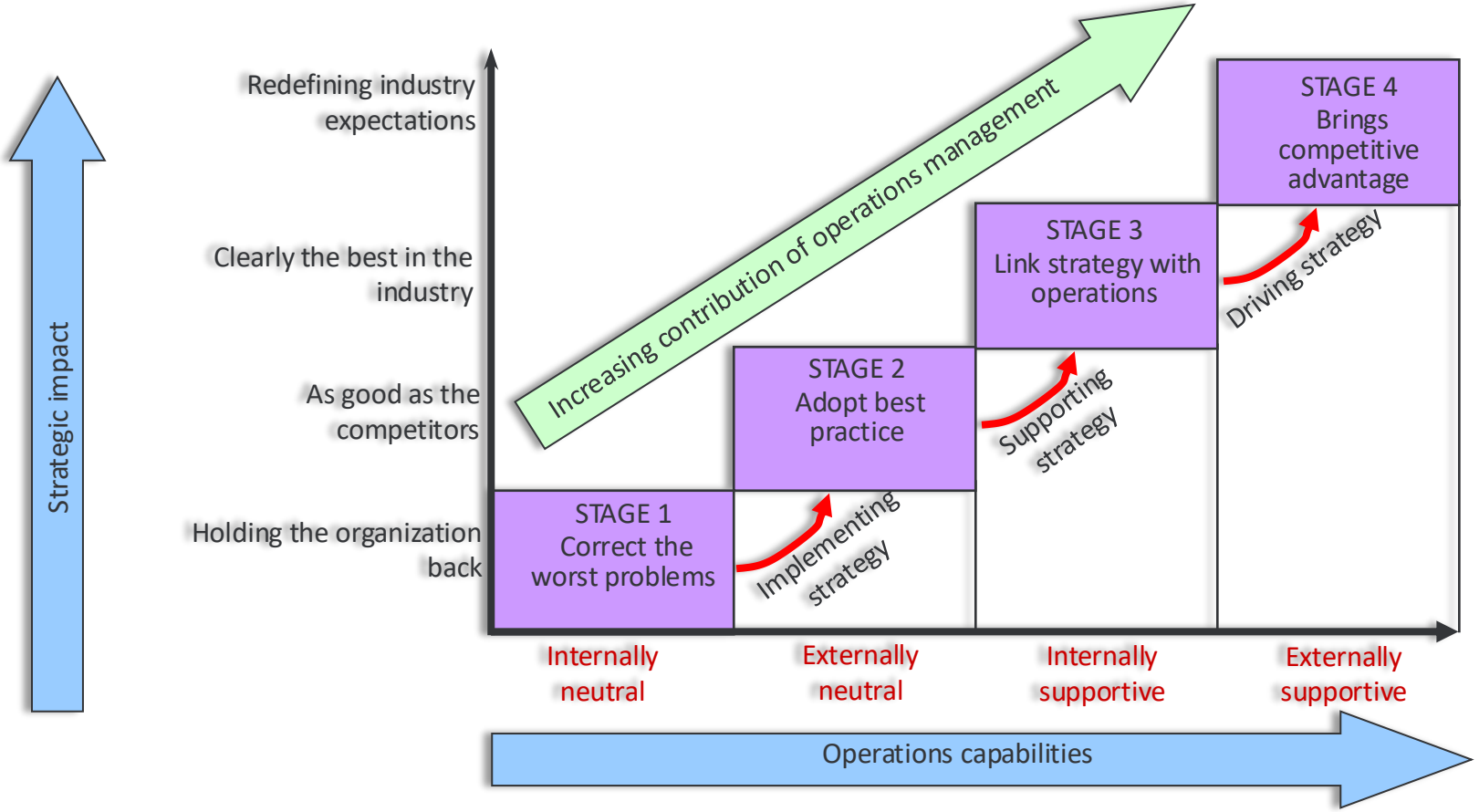
WHAT IS OPERATIONS STRATEGY?

"Operations strategy is the pattern of decisions and actions that shape the long-term vision, objectives and capabilities of the operation and its contribution to the overall strategy of the business"

Source : Brown et al. (2001)

Source: Slack et al. (2018)

THE STRATEGIC ROLE OF OM



Source: Slack et al. (2019)

COMPETITIVE AND FUNCTIONAL STRATEGY

Competitive strategy defines the set of customer needs a firm seeks to satisfy through its products and services

e.g. Walmart aims to provide high availability of a variety of products of reasonable quality at low prices

Functional strategies

OM strategy determines the nature of material procurement, transportation of materials, manufacture of product or creation of service, distribution of product

Product development strategy specifies the portfolio of new products that the company will try to develop

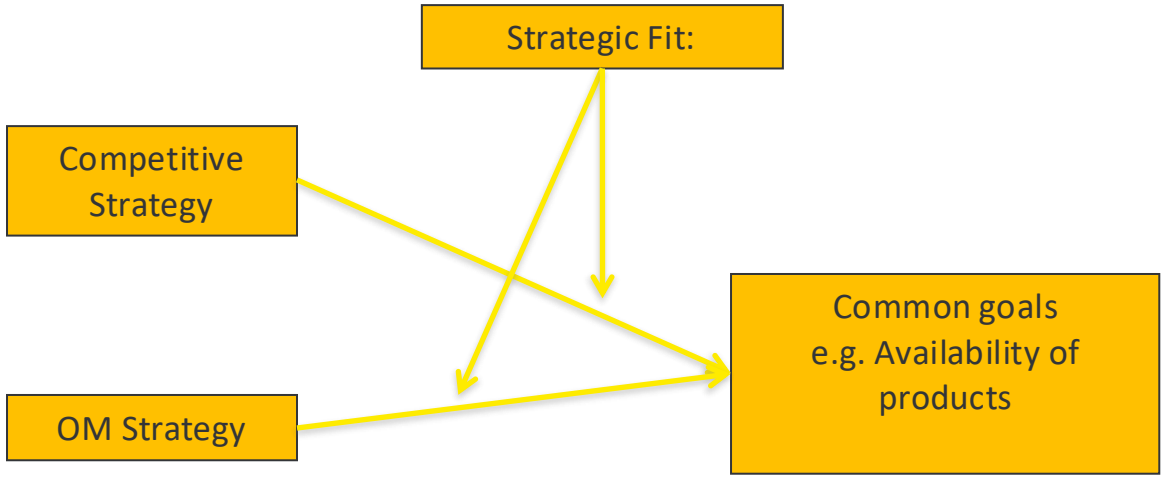
Marketing and sales strategy specifies how the market will be segmented and product positioned, priced, and promoted

"All functional strategies must support one another and the competitive strategy"

STRATEGIC FIT

Strategic fit means:

- The competitive strategy and OM strategy have aligned goals
- Processes/resources/capabilities are suitable to achieve those goals



STRATEGIC FIT: SOUTHWEST AIRLINES

- What are the main strategic objectives of Southwest Airlines?
- How does OM contribute to achieve those objectives?

OPERATIONS CAPABILITIES

- <u>Operations capabilities or Competitive priorities</u>
  - Quality
  - Delivery
  - Flexibility
  - Cost
  - Innovation*
  - Sustainability*

QUALITY

- The ability to produce unquestionable quality
- The ability to provide free-error goods that comply with specifications

Goods

Measures:
- Made to specification
- Reliable
- Consistent
- Easy to assemble and service

Services

Measures:
- Clean and tidy
- Courteous personnel, friendly and helpful

TOYOTA

Radisson

DELIVERY (SPEED & DEPENDABILITY)

- Time-based capability to deliver products at the promised time to customers
- For some customers delivery is not enough and delivery speed must be considered to win orders

Goods

Measures:
- On-time delivery of products
- On-time delivery of parts
- Dependability
- Fast lead-time

Services

Measures:
- Predictable opening hours
- Reasonable queuing times
- Stock available
- Parking spaces available

FLEXIBILITY

- The ability to adapt and respond to demand change
- Flexibility include the ability to respond to process disruptions like breakdown machines or late arrivals of raw materials

Goods

Measures:

- Fast production reaction
- Multi-purpose machines
- Multi-skilled personnel
- Fast spare parts changeovers
- Flexible suppliers

Services

Measures:

- Wide range of options
- Multi-skilled personnel

SINGAPORE AIRLINES

COST

- The ability to save resources by pursuing a given strategy
- While cost is the competitive priority of choice, considering cost as a sole performance may undermine other aspects of performance in the system

Goods

Measures:
- Staff costs
- Raw material costs
- Technology and facilities costs

Services

Measures:
- Staff costs
- Raw material costs
- Technology and facilities costs

INNOVATION

- The ability to develop new products, technologies and processes in anticipation or in response to customer requirements

Product

Measures:
- New products
- New processes
- New facilities
- New organization structures

Service

Measures:
- Innovative ways to deliver the service
- New corporate image
- New processes

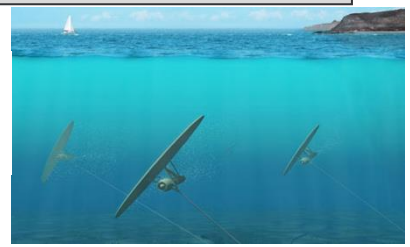
SUSTAINABILITY

- The ability to develop products that do not affect the natural and social environment while still economically feasible

Goods

Measures:
- Products that use renewable energy
- Products that encourage social responsibility practices

Minesto



Services

Measures:
- Services that encourage low CO2 emissions
- Services that improve social wellbeing

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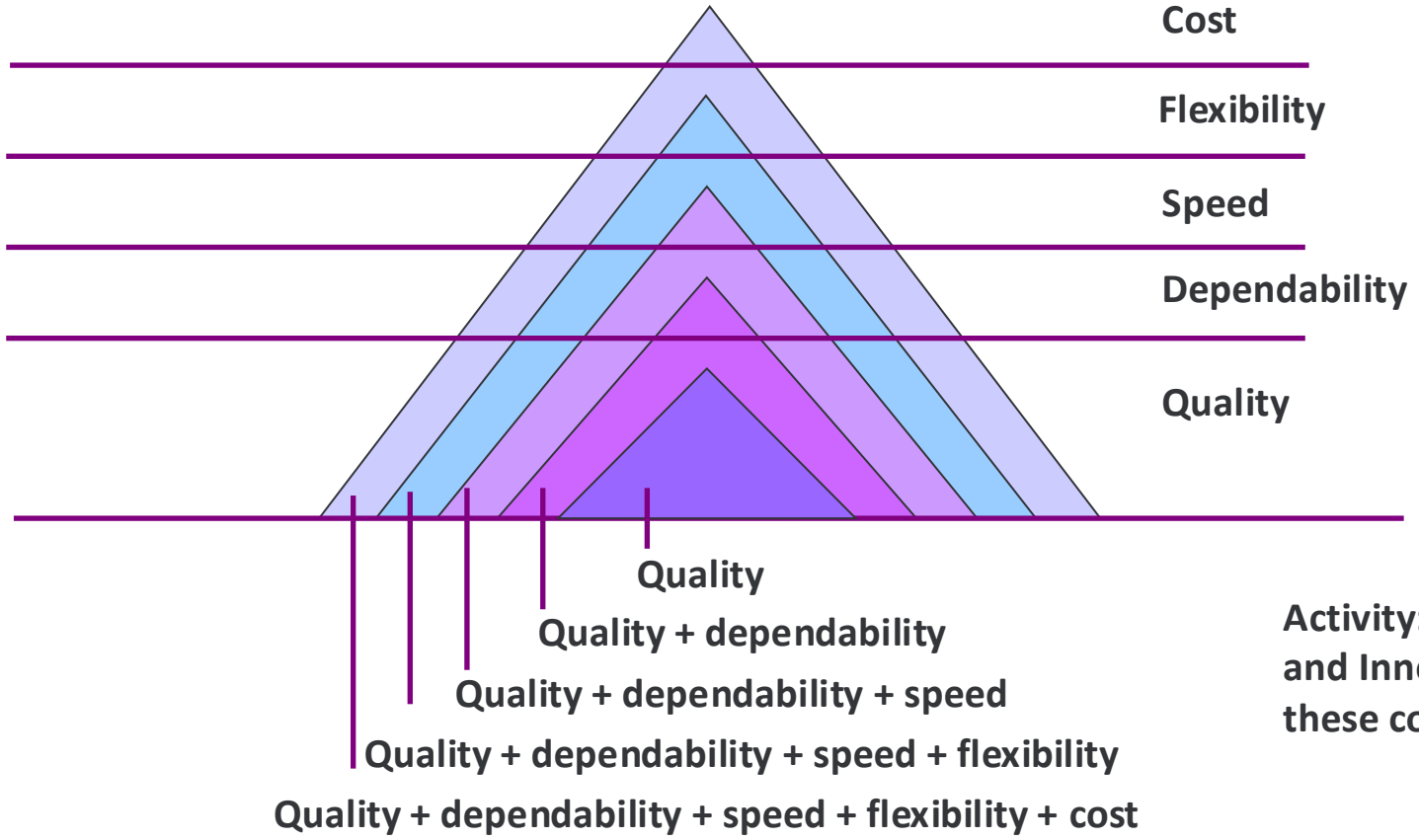
THE TRADE-OFF MODEL

- The ‘trade-off’ model suggests that there is a need to prioritise between competitive priorities
  - Companies cannot possibly compete by offering superior performance through all competitive priorities
- Managers should choose their plant’s competitive priorities first, and then design and operate the manufacturing systems that help achieve their goals

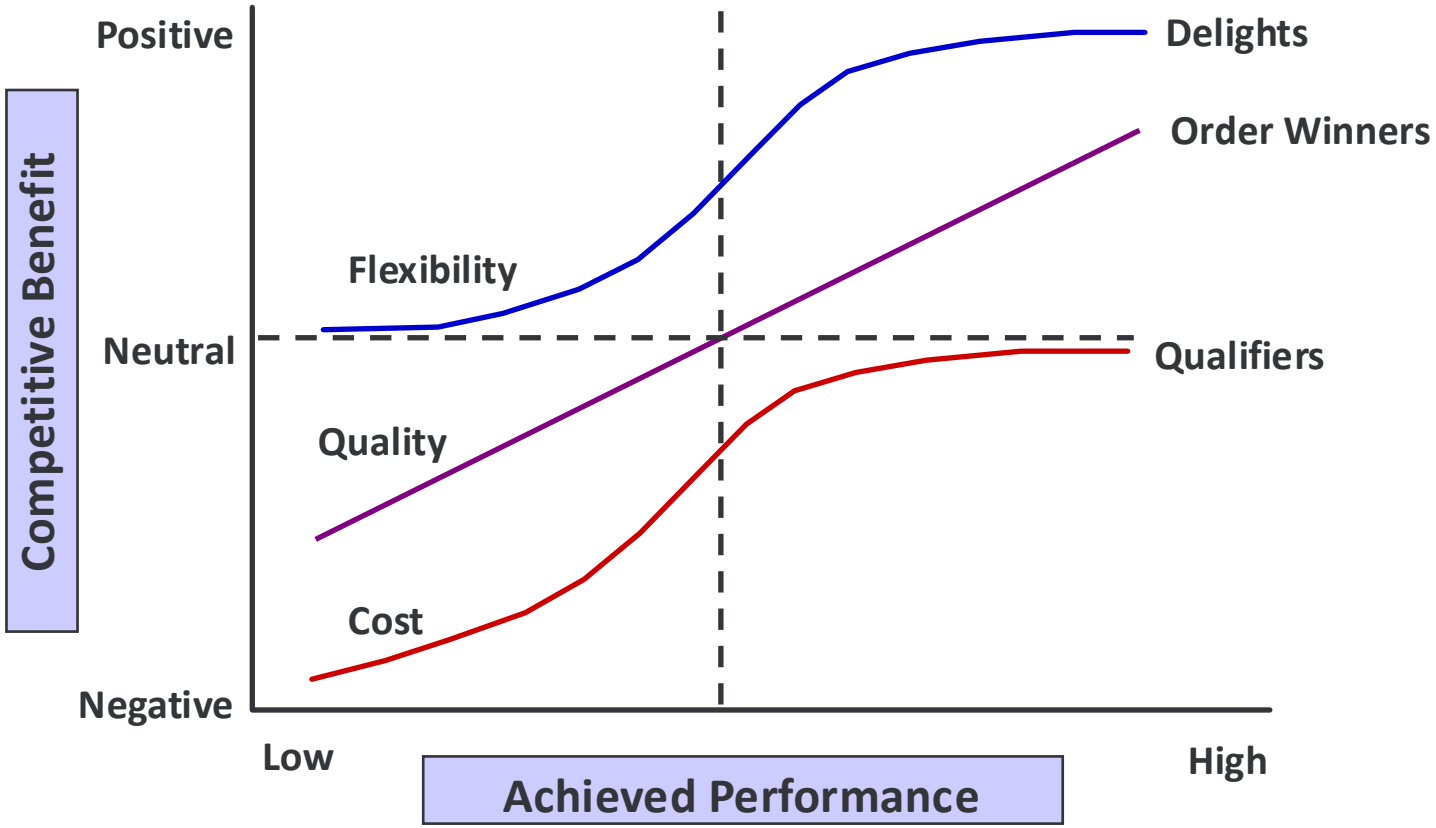
THE SANDCONE MODEL OF IMPROVEMENT; COST REDUCTION RELIES ON A CUMULATIVE FOUNDATION OF IMPROVEMENT IN THE OTHER PERFORMANCE OBJECTIVES



Activity: Regarding Sustainability and Innovation, where do you think these could fit in and how ?

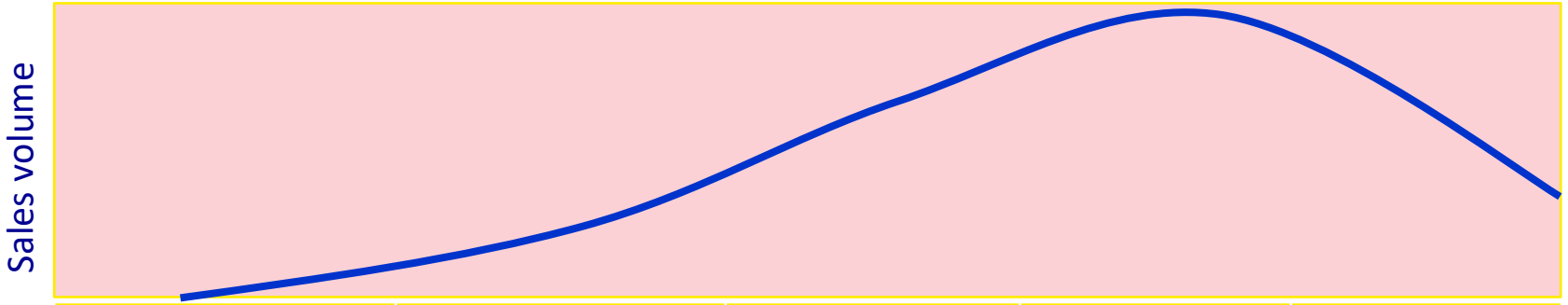
Source: Slack et al. (2018)

PRIORITIZING BETWEEN COMPETITIVE PRIORITIES: THE ORDER 'WINNERS' AND 'QUALIFIERS' MODEL



Source: Slack et al. (2018)

PRIORITIZING BETWEEN COMPETITIVE PRIORITIES: THE PRODUCT LIFE CYCLE MODEL



	Introduction	Growth	Maturity	Decline
	Product/service first introduced to market	Product/service gains market acceptance	Market needs start to be fulfilled	Market needs largely met
Volume	Slow growth in sales	Rapid growth in sales volume	Sales slow down and level off	Sales decline
Customers	Innovators	Early adopters	Bulk of market	Laggards
Competitors	Few/none	Increasing number	Stable number	Declining number
Variety of product/service designs	Possible high customization or frequent design changes	Increasingly standardized	Emerging dominant types	Possible move to commodity standardization
Likely order winners	**Product /service characteristics, performance or novelty**	**Availability of quality products/services**	**Low price, Dependable supply**	**Low price**
Likely qualifiers	**Innovation, Quality, Range**	**Price, range**	**Range, Quality**	**Dependable supply**
Dominant operations performance objectives	**Innovation, Flexibility, Quality**	**Speed, Dependability, Quality**	**Cost, Dependability**	**Cost**

THE RESOURCE-BASED VIEW

- An operations-led perspective should start with an understanding of the resource and constraints
- Tangible resources:
  - Machinery, technology and premises
- Intangible resources:
  - Relationship with suppliers,
  - knowledge in handling a process
  - Ways staff work to develop a new product
    - These intangibles are the real source of competitive advantage
    - Resource based-view (RBV)

DYNAMIC CAPABILITY VIEW

- The RBV has an inner perspective which can be extended to include external sources of competitive advantage (e.g., ERBV) (Eisenhardt, 2000)
- However, today's changing market conditions require simpler, rather than complex, capabilities
- Simple capabilities which can be easy to change or reconfigure (e.g., practices)
- Short term: Temporal advantage stems from the ability to react and change resources and capabilities
- Long term: Capabilities that are built on top of simple capabilities

DYNAMIC CAPABILITY VIEW

Can you provide an example of using dynamic capabilities to generate competitive advantage. Use a company or product/service to illustrate your answer. Work in groups.

REFERENCES

- Heizer, Jay, et al. Operations Management: Sustainability and Supply Chain Management, 2019. 13th Global Edition, Pearson Education Limited. (Chapter 1)
- Operations Management: Sustainability and Supply Chain Management, 2017, 12th Edition By Heizer, J. Render, B., Mundson, C.
- Operations Management, 2019, Ninth Edition By Slack, N, Brandon-Jones, A.
- Operations Management: Processes and Supply Chains 2018. 12th Edition By Edition, Krajewski, L.J., Malhotra, M.K, and Rizman, L.P.